INTERPOLATION (4)

DIVIDED DIFFERENCES

DEFINITION OF ~~NEWTON'S~~ DIVIDED DIFFERENCE ~~FORMULA~~:

If (x_0, y_0), (x_1, y_1), ... (x_n, y_n) be given points, The first divided difference for the argument x_0, x_1 is denoted by $[x_0, x_1]$ and is defined by $\frac{y_1 - y_0}{x_1 - x_0}$.

i.e. $$[x_0, x_1] = \frac{y_1 - y_0}{x_1 - x_0}$$

Similarly, $[x_1, x_2] = \frac{y_2 - y_1}{x_2 - x_1}$, $[x_2, x_3] = \frac{y_3 - y_2}{x_3 - x_2}$

and so on.

The second divided difference for x_0, x_1, x_2 is $$[x_0, x_1, x_2] = \frac{[x_1, x_2] - [x_0, x_1]}{x_2 - x_0}$$

similarly $$[x_1, x_2, x_3] = \frac{[x_2, x_3] - [x_1, x_2]}{x_3 - x_1}$$

The third divided difference for x_0, x_1, x_2, x_3 is

$$[x_0, x_1, x_2, x_3] = \frac{[x_1, x_2, x_3] - [x_0, x_1, x_2]}{x_3 - x_0}$$

$$[x_1, x_2, x_3, x_4] = \frac{[x_2, x_3, x_4] - [x_1, x_2, x_3]}{x_4 - x_1}$$

NEWTON'S DIVIDED DIFFERENCE FORMULA

We consider the following data set

x	x_0	x_1	x_2	....	x_n
$f(x)$	$f(x_0)$	$f(x_1)$	$f(x_2)$		$f(x_n)$

Let us assume that $f(x)$ is ~~lear~~ linear.

Then $\dfrac{f(x_i) - f(x_j)}{x_i - x_j}$ (where x_i and x_j are any two tabular points) is independent of x_i and x_j.

This ratio is called the first divided difference of $f(x)$ relative to x_i and x_j and is denoted by $f[x_i, x_j]$ i.e.,

$$f[x_i, x_j] = \frac{f(x_i) - f(x_j)}{x_i - x_j} = f[x_j, x_i]$$

Since the ratio is independent of x_i and x_j, we can write $f[x_0, x] = f[x_0, x_1]$

$$\Rightarrow \frac{f(x) - f(x_0)}{x - x_0} = f[x_0, x_1]$$

$$\Rightarrow f(x) = f(x_0) + (x - x_0) f[x_0, x_1]$$

So, if $f(x)$ is approximated with a linear polynomial then the function value at any point x can be calculated as

$$f(x) \simeq P_1(x) = f(x_0) + (x - x_0) f[x_0, x_1].$$

If $f(x)$ is a 2nd degree polynomial, Then the secant slope is not constant, but a linear function of x.

Hence we have, $\dfrac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$ is independent of x_0, x_1 and x_2.

This ratio is defined as second divided difference of f relative to x_0, x_1, x_2.

$$\therefore f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$$

Since $f[x_0, x_1, x_2]$ is independent of x_0, x_1, x_2

So $f[x_0, x_1, x] = f[x_0, x_1, x_2]$

$$\Rightarrow \frac{f[x_0, x] - f[x_1, x_1]}{x - x_1} = f[x_0, x_1, x_2]$$

$$\Rightarrow f[x_0, x] = f[x_0, x_1] + (x - x_1) f[x_0, x_1, x_2]$$

$$\Rightarrow \frac{f(x) - f(x_0)}{x - x_0} = f[x_0, x_1] + (x - x_1) f[x_0, x_1, x_2]$$

$$\Rightarrow f(x) = f(x_0) + (x-x_0)\, f[x_0, x_1] + (x-x_0)(x-x_1)\, f[x_0, x_1, x_2]$$

In the same way if we define recursively kth divided difference by the relation

$$f[x_0, x_1, x_2, \ldots, x_k] = \frac{f[x_1, x_2, \ldots, x_k] - f[x_0, x_1, \ldots, x_{k-1}]}{x_k - x_0}$$

So the kth degree polynomial approximation to $f(x)$ can be written as

$$f(x) = f(x_0) + (x-x_0)\, f[x_0, x_1] + (x-x_0)(x-x_1)\, f[x_0, x_1, x_2]$$

$$+ \cdots\cdots + (x-x_0)(x-x_1)\cdots(x-x_{k-1})\, f[x_0, x_1, \ldots, x_k]$$

$$= f(x_0) + \sum_{k=1}^{n} (x-x_0)(x-x_1)\cdots(x-x_{k-1})\, f[x_0, x_1, \ldots, x_k]$$

This formula is known as Newton's interpolatory divided difference formula.

DIVIDED DIFFERENCE TABLE:

x	$f(x)$	1st Divided difference	2nd divided difference	3rd divided Difference
x_0	$f[x_0]$			
		$f[x_0,x_1]=\frac{f[x_1]-f[x_0]}{x_1-x_0}$		
x_1	$f[x_1]$		$f[x_0,x_1,x_2]=\frac{f[x_1,x_2]-f[x_0,x_1]}{x_2-x_0}$	
		$f[x_1,x_2]=\frac{f[x_2]-f[x_1]}{x_2-x_1}$		$f[x_0,x_1,x_2,x_3]=\frac{f[x_1,x_2,x_3]-f[x_0,x_1,x_2]}{x_3-x_0}$
x_2	$f[x_2]$		$f[x_1,x_2,x_3]=\frac{f[x_2,x_3]-f[x_1,x_2]}{x_3-x_2}$	
		$f[x_2,x_3]=\frac{f[x_3]-f[x_2]}{x_3-x_2}$		$f[x_1,x_2,x_3,x_4]=\frac{f[x_2,x_3,x_4]-f[x_1,x_2,x_3]}{x_4-x_1}$
x_3	$f[x_3]$		$f[x_2,x_3,x_4]=\frac{f[x_3,x_4]-f[x_2,x_3]}{x_3-x_2}$	
		$f[x_3,x_4]=\frac{f[x_4]-f[x_3]}{x_4-x_3}$		$f[x_2,x_3,x_4,x_5]=\frac{f[x_3,x_4,x_5]-f[x_2,x_3,x_4]}{x_5-x_2}$
x_4	$f[x_4]$		$f[x_3,x_4,x_5]=\frac{f[x_4,x_5]-f[x_3,x_4]}{x_5-x_4}$	
		$f[x_4,x_5]=\frac{f[x_5]-f[x_4]}{x_5-x_4}$		
x_5	$f[x_5]$			

Example: The following table gives the values of a function at various points. Find $f(1.5)$ by divided difference formula.

x	1	1.3	1.6	1.9	2.2
$f(x)$	0.7651977	0.620086	0.4554022	0.2818186	0.1103623

Solution: The difference table is as follows:

x_i	$f(x)=f[x_i]$	$f[x_{i-1},x_i]$	$f[x_{i-2},x_{i-1},x_i]$	$f[x_{i-3},\dots,x_i]$	$f[x_{i-4},\dots,x_i]$
$x_0=1.0$	0.7651977				
		−0.4837057			
$x_1=1.3$	0.620086		−0.1087339		
		−0.5489460		0.0658784	
$x_2=1.6$	0.4554022		−0.0494433		0.0018251
		−0.5786120		0.0680685	
$x_3=1.9$	0.2818186		0.0118183		
		−0.5715210			
$x_4=2.2$	0.1103623				

We know,
$$P(x)=f(x_0)+(x-x_0)f[x_0,x_1]+(x-x_0)(x-x_1)f[x_0,x_1,x_2]+(x-x_0)(x-x_1)(x-x_2)f[x_0,x_1,x_2,x_3]+(x-x_0)(x-x_1)(x-x_2)(x-x_3)f[x_0,x_1,\dots,x_4]$$

$$\Rightarrow P(x)=0.7651977+(x-1)\times(-0.4837057)-0.1087339(x-1)(x-1.3)+0.0658784(x-1)(x-1.3)(x-1.6)+0.0018251(x-1)(x-1.3)(x-1.6)(x-1.9)$$

$$\boxed{\therefore P(1.5)=0.5118200}$$